Amortization gaps of fixed encoder classes: an N-independent upper bound and a Lipschitz lower bound

Draft: notation, Lemma 1, Theorems 3 and 7

Abstract

We work in a simple hierarchical model with a point-estimate decoder. Theorem 3 shows that if the decoder fitted by factorized variational inference (F-VI) stabilizes as the sample size N grows, the per-datapoint amortization gap of any fixed encoder class is asymptotically at most a population quantity that does not depend on N. The bound holds for every N with explicit deviation terms and needs no condition on the size of the class. Theorem 7 shows, at a fixed decoder, that when the optimal variational parameters jump between separated basins, every Λ-Lipschitz encoder pays a per-datapoint gap at least $\min(c/\Lambda, c')$, where c is set by the height of the barrier between the basins and c' by the cost of the wrong basin. This bound also holds for the training gap, uniformly over all Λ-Lipschitz encoders. Together, the two results say that the size of an encoder need not grow with N, but that its Lipschitz constant must grow as $1/\varepsilon$ to reach a per-datapoint gap ε when the optimal map jumps. We relate both results to Margossian and Blei (2024), to Cremer et al. (2018), and to work on Lipschitz maps and multimodal distributions.[1]

1 Setting and notation

Model. We consider a simple hierarchical model with decoder parameters θ and one local latent variable per datapoint,

$$p_\theta(z, x) = \prod_{n=1}^{N} p(z_n)\, p_\theta(x_n \mid z_n)$$

and data $x = (x_1, \ldots, x_N) \in \mathcal{X}^N$. The decoder is a point estimate (variational EM); for simplicity the prior on θ is flat.[2] We write $\log p_\theta(x) = \sum_n \log p_\theta(x_n)$.

Classes.

- Θ: the class of decoders (an architecture), the same for every method.
- $\mathcal{Q} = \{q(\cdot, \nu) : \nu \in U\}$: the variational family (e.g. FFG or a flow), with parameter space U, an open subset of a Euclidean space.
- $\mathcal{F}$: the class of encoders, i.e. maps $f : \mathcal{X} \to U$. The family $\mathcal{Q}$ is implied by the codomain of $\mathcal{F}$, and F-VI optimizes over all of U.

[1] Theorems 3 and 7 are Theorems 3 and 7 of the working notes (v5), restated in the notation of the gap draft. In this point-estimate setting, the concentration of the global variational factor is replaced by the convergence of the F-VI decoder; the Bayesian version is Remark 8.

[2] With a prior $p(\theta)$, add $\log p(\theta)$ to $\log p_\theta(x)$, $\mathcal{L}_F(\theta, \mathcal{Q})$ and $\mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q})$ below; nothing else changes.

Objectives. The ELBO of datapoint n for parameters ν, decoder θ and a variational family $\mathcal{Q}$:

$$\begin{aligned}\mathcal{L}_n(\nu, \theta, \mathcal{Q}) &= \mathbb{E}_{q(z,\nu)}\big[\log p(z) + \log p_\theta(x_n \mid z) - \log q(z, \nu, \mathcal{Q})\big] \\ &= \log p_\theta(x_n) - \mathrm{KL}\big(q(\cdot, \nu, \mathcal{Q}) \,\|\, p_\theta(\cdot \mid x_n)\big).\end{aligned}$$

For a decoder θ and a variational family $\mathcal{Q}$, the best ELBO of F-VI and of A-VI are

$$\mathcal{L}_F(\theta, \mathcal{Q}) \coloneqq \sup_{\nu_1,\dots,\nu_N \in U} \sum_{n=1}^N \mathcal{L}_n(\nu_n, \theta, \mathcal{Q}), \qquad \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) \coloneqq \sup_{f \in \mathcal{F}} \sum_{n=1}^N \mathcal{L}_n\big(f(x_n), \theta, \mathcal{Q}\big).$$

The per-datapoint suboptimality of ν is

$$\delta_n(\nu, \theta, \mathcal{Q}) \coloneqq \sup_{\nu' \in U} \mathcal{L}_n(\nu', \theta, \mathcal{Q}) - \mathcal{L}_n(\nu, \theta, \mathcal{Q}) \geq 0.$$

Separation of the ELBO. For a factorized distribution $q(z) = \prod_{n=1}^N q(z_n, \nu_n, \mathcal{Q})$, the ELBO of the whole dataset is

$$\mathcal{L}(\nu_{1:N}, \theta, \mathcal{Q}) \coloneqq \mathbb{E}_{q(z)}\big[\log p_\theta(z, x) - \log q(z)\big].$$

F-VI maximizes it over $\nu_{1:N} \in U^N$; A-VI maximizes it over $\nu_n = f(x_n)$, $f \in \mathcal{F}$. In a simple hierarchical model,

$$\mathcal{L}(\nu_{1:N}, \theta, \mathcal{Q}) = \sum_{n=1}^N \mathcal{L}_n(\nu_n, \theta, \mathcal{Q}), \tag{1}$$

$$\mathcal{L}_F(\theta, \mathcal{Q}) = \sup_{\nu_{1:N} \in U^N} \mathcal{L}(\nu_{1:N}, \theta, \mathcal{Q}) = \sum_{n=1}^N \sup_{\nu \in U} \mathcal{L}_n(\nu, \theta, \mathcal{Q}), \tag{2}$$

and likewise $\mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) = \sup_{f \in \mathcal{F}} \mathcal{L}\big(f(x_1), \dots, f(x_N), \theta, \mathcal{Q}\big)$. In words: the ELBO is a sum of per-datapoint ELBOs, each depending on its own ν_n only, so the F-VI maximization splits into N independent maximizations.

Proof of (1) and (2). Log densities are sums. By the model and the factorization of q,

$$\log p_\theta(z, x) = \sum_{n=1}^N \big[\log p(z_n) + \log p_\theta(x_n \mid z_n)\big], \qquad \log q(z) = \sum_{n=1}^N \log q(z_n, \nu_n, \mathcal{Q}).$$

The n-th term of each sum depends on z_n only.

Expectations of local terms. By linearity,

$$\mathcal{L}(\nu_{1:N}, \theta, \mathcal{Q}) = \sum_{n=1}^N \mathbb{E}_{q(z)}\big[\log p(z_n) + \log p_\theta(x_n \mid z_n) - \log q(z_n, \nu_n, \mathcal{Q})\big].$$

The n-th integrand is a function $h_n(z_n)$. Since q is a product and every factor $q(z_m, \nu_m, \mathcal{Q})$ integrates to one, integrating out z_m, $m \neq n$ (Fubini) gives $\mathbb{E}_{q(z)}[h_n(z_n)] = \mathbb{E}_{q(z_n,\nu_n,\mathcal{Q})}[h_n(z_n)] = \mathcal{L}_n(\nu_n, \theta, \mathcal{Q})$. This is (1).

The supremum of a separated sum. By (1), $\mathcal{L}$ is a sum of functions $\mathcal{L}_n(\cdot, \theta, \mathcal{Q})$, each of its own variable ν_n. For every $\nu_{1:N}$, $\sum_n \mathcal{L}_n(\nu_n, \theta, \mathcal{Q}) \leq \sum_n \sup_\nu \mathcal{L}_n(\nu, \theta, \mathcal{Q})$. Conversely, assuming the suprema are finite, for every $\varepsilon > 0$ choose ν_n with $\mathcal{L}_n(\nu_n, \theta, \mathcal{Q}) \geq \sup_\nu \mathcal{L}_n(\nu, \theta, \mathcal{Q}) - \varepsilon/N$; then $\sum_n \mathcal{L}_n(\nu_n, \theta, \mathcal{Q}) \geq \sum_n \sup_\nu \mathcal{L}_n(\nu, \theta, \mathcal{Q}) - \varepsilon$. This is (2).

Second form of $\mathcal{L}_n$. Writing $\log p(z) + \log p_\theta(x_n \mid z) = \log p_\theta(x_n) + \log p_\theta(z \mid x_n)$,

$$\begin{aligned}\mathcal{L}_n(\nu, \theta, \mathcal{Q}) &= \log p_\theta(x_n) + \mathbb{E}_{q(z,\nu)}\big[\log p_\theta(z \mid x_n) - \log q(z, \nu, \mathcal{Q})\big] \\ &= \log p_\theta(x_n) - \mathrm{KL}\big(q(\cdot, \nu, \mathcal{Q}) \,\|\, p_\theta(\cdot \mid x_n)\big).\end{aligned}$$

□

Remark 1 (what separation needs). Two structural assumptions are used: each term of $\log p_\theta(z, x)$ involves a single z_n (simple hierarchical model), and q is a product over datapoints. In a hidden Markov model, $\log p(z_n \mid z_{n-1})$ couples z_n and z_{n-1}, so its expectation depends on (ν_n, ν_{n-1}): (1) fails and the optimal ν_n depends on its neighbours.

Decoders. We assume the following maximizers exist in Θ and the maxima are finite:

$$\hat{\theta}_{\mathrm{ML}} \in \arg\max_{\theta\in\Theta} \log p_\theta(x), \qquad \hat{\theta}_F \in \arg\max_{\theta\in\Theta} \mathcal{L}_F(\theta, \mathcal{Q}), \qquad \hat{\theta}_A \in \arg\max_{\theta\in\Theta} \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}).$$

$\hat{\theta}_A$ is the decoder of a VAE trained to its optimum, $\hat{\theta}_F$ that of variational EM with free local parameters. If the maxima are not attained, the statements below hold up to ε with ε-maximizers. The maximizers over all decoders, $\theta^*_{\mathrm{ML}}, \theta^*_F, \theta^*_A$, are the special case where Θ is unconstrained, whenever they exist.

Definition 1 (Amortization gaps)**.** For a decoder $\theta \in \Theta$ (Θ represents the decoder class) and a variational family $\mathcal{Q}$:

$$\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q}) \coloneqq \mathcal{L}_F(\theta, \mathcal{Q}) - \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}),$$

$$\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) \coloneqq \sup_{\theta\in\Theta} \mathcal{L}_F(\theta, \mathcal{Q}) - \sup_{\theta\in\Theta} \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_A(\hat{\theta}_A, \mathcal{F}, \mathcal{Q}).$$

$\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})$ is the amortization gap at a fixed decoder; it is the quantity measured by Cremer et al. (2018) at $\theta = \hat{\theta}_A$. $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q})$ compares the best F-VI and the best A-VI objectives, each with its own decoder, as described by Margossian and Blei (2024). When $\Theta = \{\theta\}$ (i.e. the decoder is constrained) the two notations match: $\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q}) = \mathcal{G}_{\mathrm{am}}(\{\theta\}, \mathcal{F}, \mathcal{Q})$.

2 The gap identity

Lemma 1. *In a simple hierarchical model, for every class $\mathcal{F}$ and $\mathcal{Q}$:*

(i) (fixed decoder) *for every $\theta \in \Theta$,*

$$\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q}) = \inf_{f\in\mathcal{F}} \sum_{n=1}^{N} \delta_n\big(f(x_n), \theta, \mathcal{Q}\big);$$

(ii) (decoder class)

$$\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) = \inf_{\theta\in\Theta} \Big[\big(\mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_F(\theta, \mathcal{Q})\big) + \mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})\Big],$$

and the infimum is attained at $\theta = \hat{\theta}_A$.

Proof. (i) For $f \in \mathcal{F}$, by (2),

$$\begin{aligned}\mathcal{L}_F(\theta, \mathcal{Q}) - \sum_n \mathcal{L}_n(f(x_n), \theta, \mathcal{Q}) &= \sum_n \Big[\sup_\nu \mathcal{L}_n(\nu, \theta, \mathcal{Q}) - \mathcal{L}_n(f(x_n), \theta, \mathcal{Q})\Big] \\ &= \sum_n \delta_n(f(x_n), \theta, \mathcal{Q}).\end{aligned}$$

Taking the supremum over f of $\sum_n \mathcal{L}_n(f(x_n), \theta, \mathcal{Q})$ gives (i).

(ii) For every θ, $\mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_F(\theta, \mathcal{Q}) - \mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})$, so

$$\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \sup_\theta \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) = \inf_\theta \big[\mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_F(\theta, \mathcal{Q}) + \mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})\big].$$

At $\theta = \hat{\theta}_A$ the bracket equals $\mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_A(\hat{\theta}_A, \mathcal{F}, \mathcal{Q}) = \mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q})$. □

Corollary 2 (two inequalities)**.**

$$\mathcal{G}_{\mathrm{am}}(\hat{\theta}_A, \mathcal{F}, \mathcal{Q}) \le \mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) \le \mathcal{G}_{\mathrm{am}}(\hat{\theta}_F, \mathcal{F}, \mathcal{Q}).$$

Proof. Expand the definitions: $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) - \mathcal{G}_{\mathrm{am}}(\hat{\theta}_A, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_F(\hat{\theta}_A, \mathcal{Q})$, nonnegative since $\hat{\theta}_F$ maximizes $\mathcal{L}_F$, and $\mathcal{G}_{\mathrm{am}}(\hat{\theta}_F, \mathcal{F}, \mathcal{Q}) - \mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_A(\hat{\theta}_A, \mathcal{F}, \mathcal{Q}) - \mathcal{L}_A(\hat{\theta}_F, \mathcal{F}, \mathcal{Q})$, nonnegative since $\hat{\theta}_A$ maximizes $\mathcal{L}_A$. □

Remark 2. The left inequality says that the amortization gap measured at the decoder of the trained VAE underestimates $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q})$, by the loss of F-VI when it is restricted to that decoder. The right inequality is part (ii) of Lemma 1 evaluated at $\theta = \hat{\theta}_F$. With $\Theta = \{\theta\}$, all three quantities coincide.

Remark 3 (status). Lemma 1 is partially known. In variational EM, Shu et al. (2018, Sec. 3, Eqs. 11–12) write the unamortized and the amortized objectives as the expected log-likelihood minus $\mathbb{E} \min_q \mathrm{KL}(q \,\|\, p_\theta(\cdot \mid x))$ and minus $\min_f \mathbb{E}\, \mathrm{KL}(f(x) \,\|\, p_\theta(\cdot \mid x))$; the difference of their optimal values is (ii). Part (i) is the amortization gap of Cremer et al. (2018, Sec. 3.1, Eq. 8) summed over datapoints.

3 A fixed-size encoder suffices as N grows

Interpolating arbitrary real targets on N points requires a class of pseudo-dimension at least N; for ReLU networks with W weights and L layers this means $WL \log W \gtrsim N$ (Bartlett et al., 2019). An encoder that had to interpolate the F-VI optima would therefore grow with N. It does not have to, because the targets are not arbitrary. In a simple hierarchical model they depend on the whole dataset only through the decoder, and once the F-VI decoder stabilizes they become values of a single limiting function, which a fixed class only has to approximate on average. This section makes this precise.

Conventions. From now on x denotes a single observation, an element of $\mathcal{X}$, and $x_{1:N} \in \mathcal{X}^N$ the dataset; we write $\log p_\theta(x_{1:N}) = \sum_n \log p_\theta(x_n)$. $\hat{\theta}_F$, $\hat{\theta}_A$ and the gaps depend on N and on $x_{1:N}$; we keep the notation of Section 1 without an index N.

3.1 Notation for the limiting problem

ELBO at a generic observation. The limit evaluates the gap at an observation that is not in the dataset, so we index the per-datapoint quantities by the observation rather than by n. For $x \in \mathcal{X}$,

$$\mathcal{L}_x(\nu, \theta, \mathcal{Q}) \coloneqq \mathbb{E}_{q(z,\nu)}\big[\log p(z) + \log p_\theta(x \mid z) - \log q(z, \nu, \mathcal{Q})\big],$$
$$\delta_x(\nu, \theta, \mathcal{Q}) \coloneqq \sup_{\nu' \in U} \mathcal{L}_x(\nu', \theta, \mathcal{Q}) - \mathcal{L}_x(\nu, \theta, \mathcal{Q}).$$

Thus $\mathcal{L}_n = \mathcal{L}_{x_n}$ and $\delta_n = \delta_{x_n}$. Whenever $\log p_\theta(x)$ is finite,

$$\delta_x(\nu, \theta, \mathcal{Q}) = \mathrm{KL}\big(q(\cdot, \nu, \mathcal{Q}) \,\|\, p_\theta(\cdot \mid x)\big) - \inf_{\nu' \in U} \mathrm{KL}\big(q(\cdot, \nu', \mathcal{Q}) \,\|\, p_\theta(\cdot \mid x)\big)$$

is an excess KL divergence.

Data, metric, range of the encoders. P is the distribution of the observations; it is needed to let N grow and to define new data. d is a metric on Θ, needed to speak of convergence of the decoder. $K \subset U$ is the set in which the encoders take their values, in practice a compact box (e.g. bounded means and log-standard deviations for a factorized Gaussian family); uniform controls hold only on such a set.

Limit decoder, modulus of continuity, bound. $\theta^\dagger \in \Theta$ denotes the limit of $\hat{\theta}_F$ (Assumption (H2) below). The modulus of continuity of the ELBO at $\theta^\dagger$, uniform over $K \times \mathcal{X}$, is

$$\omega(r) \coloneqq \sup_{\theta\in\Theta:\ d(\theta,\theta^\dagger)\le r} \ \sup_{(\nu,x)\in K\times\mathcal{X}} \left|\mathcal{L}_x(\nu,\theta,\mathcal{Q}) - \mathcal{L}_x(\nu,\theta^\dagger,\mathcal{Q})\right| \ \in [0,\infty].$$

Since $\log p(z)$ and $\log q(z,\nu,\mathcal{Q})$ do not depend on θ, the difference equals $\mathbb{E}_{q(z,\nu)}[\log p_\theta(x \mid z) - \log p_{\theta^\dagger}(x \mid z)]$: only the reconstruction term enters. ω is the price of replacing the data-dependent decoder $\hat{\theta}_F$ by the fixed decoder $\theta^\dagger$, and it makes the bound quantitative. Finally,

$$B \coloneqq \sup_{(\nu,x)\in K\times\mathcal{X}} \delta_x(\nu,\theta^\dagger,\mathcal{Q})$$

bounds the per-datapoint suboptimality at $\theta^\dagger$; Hoeffding's inequality needs it.

Definition 2 (Population amortization gap)**.** For $\theta \in \Theta$ and a class $\mathcal{F}$ of measurable maps $\mathcal{X} \to K$,

$$\mathcal{R}(\theta,\mathcal{F},\mathcal{Q}) \coloneqq \inf_{f\in\mathcal{F}} \ \mathbb{E}_{X\sim P}\big[\delta_X\big(f(X),\theta,\mathcal{Q}\big)\big].$$

By Lemma 1(i), $\mathcal{G}_{\mathrm{am}}(\theta,\mathcal{F},\mathcal{Q})/N = \inf_{f\in\mathcal{F}} \frac{1}{N}\sum_n \delta_{x_n}\big(f(x_n),\theta,\mathcal{Q}\big)$; $\mathcal{R}$ is the same quantity with the empirical mean replaced by the expectation under P. It is what the best encoder of $\mathcal{F}$ loses per datapoint, on average over new data, at decoder θ: an expected excess KL divergence, which does not depend on N.

3.2 Assumptions

(H1) Data. $x_1, x_2, \ldots$ are i.i.d. with distribution P on $\mathcal{X}$.

(H2) Stabilization of the F-VI decoder. For each N, $\hat{\theta}_F$ maximizes $\mathcal{L}_F(\cdot,\mathcal{Q})$ over Θ (Section 1), and $d(\hat{\theta}_F,\theta^\dagger) \to 0$ in probability as $N \to \infty$.

(H3) Per-datapoint optima stay in K. There is $r_0 > 0$ such that, for every $\theta \in \Theta$ with $d(\theta,\theta^\dagger) < r_0$ and every $x \in \mathcal{X}$, $\sup_{\nu\in U} \mathcal{L}_x(\nu,\theta,\mathcal{Q}) = \sup_{\nu\in K} \mathcal{L}_x(\nu,\theta,\mathcal{Q})$. The targets of the encoder stay in the region where the encoders live and where ω controls the ELBO.

(H4) Continuity in the decoder. $\omega(r) \to 0$ as $r \to 0$.

(H5) Bounded suboptimality at $\theta^\dagger$. $\mathcal{L}_x(\nu,\theta^\dagger,\mathcal{Q})$ is finite on $K \times \mathcal{X}$, $(\nu,x) \mapsto \delta_x(\nu,\theta^\dagger,\mathcal{Q})$ is measurable, and $B < \infty$.

If $\mathcal{X}$ and K are compact, Θ is a closed subset of a Euclidean space, $(\nu,x,\theta) \mapsto \mathcal{L}_x(\nu,\theta,\mathcal{Q})$ is continuous and (H3) holds, then (H4) and (H5) hold by uniform continuity on compact sets; compactness serves no other purpose. We assume that $\hat{\theta}_F$ and the gaps are measurable. Otherwise Pr is replaced by the outer probability, and the proof is unchanged because it uses only the monotonicity and subadditivity of Pr.

3.3 Statement

Theorem 3 (N-independent bound)**.** *Assume (H1)–(H5), and let $\mathcal{F}$ be a nonempty class of measurable maps $\mathcal{X} \to K$ that does not depend on N; no other condition is imposed on $\mathcal{F}$.* Pr *is the distribution of $x_{1:N}$.*

(i) For every $N \ge 1$, every encoder $f_0 \in \mathcal{F}$ fixed in advance (not depending on the data), every $r \in (0, r_0)$ and every $\alpha \in (0,1)$, with probability at least $1 - \alpha - \Pr\big(d(\hat{\theta}_F,\theta^\dagger) \ge r\big)$,

$$\frac{\mathcal{G}_{\mathrm{am}}(\hat{\theta}_F,\mathcal{F},\mathcal{Q})}{N} \ \le \ \mathbb{E}_{X\sim P}\big[\delta_X\big(f_0(X),\theta^\dagger,\mathcal{Q}\big)\big] + 2\,\omega(r) + B\sqrt{\frac{\log(1/\alpha)}{2N}}. \tag{3}$$

(ii) For every $\varepsilon > 0$ and every $r \in (0, r_0)$ with $2\,\omega(r) \le \varepsilon/3$ (such r exist by (H4)),

$$\Pr\Big(\frac{\mathcal{G}_{\mathrm{am}}(\hat\theta_F, \mathcal{F}, \mathcal{Q})}{N} > \mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) + \varepsilon\Big) \le \Pr\big(d(\hat\theta_F, \theta^\dagger) \ge r\big) + \exp\Big(-\frac{2N\varepsilon^2}{9B^2}\Big), \quad (4)$$

and the right-hand side tends to 0 as $N \to \infty$.

(iii) (i) and (ii) hold with $\mathcal{G}_{\mathrm{am}}(\hat\theta_F, \mathcal{F}, \mathcal{Q})$ replaced by the global gap $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q})$ or by the gap at the VAE decoder, $\mathcal{G}_{\mathrm{am}}(\hat\theta_A, \mathcal{F}, \mathcal{Q})$.

If $B = 0$, the last term of (3) and the exponential term of (4) vanish.

Reading. The bound concerns the amortization gap per training datapoint at the F-VI decoder,

$$\frac{\mathcal{G}_{\mathrm{am}}(\hat\theta_F, \mathcal{F}, \mathcal{Q})}{N} = \frac{\mathcal{L}_F(\hat\theta_F, \mathcal{Q}) - \mathcal{L}_A(\hat\theta_F, \mathcal{F}, \mathcal{Q})}{N}.$$

By (3) it is at most the sum of three terms: what f_0 loses on average over new data at the limit decoder $\theta^\dagger$ (at best $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q})$); the price $2\,\omega(r)$ of an F-VI decoder that has not reached its limit; and the fluctuation of a mean of N bounded i.i.d. terms. None of these grows with N, and the last two vanish. For large N, a class of fixed size therefore keeps the per-datapoint gap below $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) + \varepsilon$.

3.4 Proof

Proof of Theorem 3. Step 1: a single encoder chosen in advance. For every dataset and every $f_0 \in \mathcal{F}$, Lemma 1(i) at $\theta = \hat\theta_F$ gives

$$\mathcal{G}_{\mathrm{am}}(\hat\theta_F, \mathcal{F}, \mathcal{Q}) = \inf_{f \in \mathcal{F}} \sum_{n=1}^{N} \delta_n\big(f(x_n), \hat\theta_F, \mathcal{Q}\big) \le \sum_{n=1}^{N} \delta_n\big(f_0(x_n), \hat\theta_F, \mathcal{Q}\big). \quad (5)$$

The right-hand side is not a sum of independent variables: $\hat\theta_F$ depends on all the data, hence so does each term. Step 2 removes this dependence.

Step 2: from $\hat\theta_F$ to $\theta^\dagger$. We show that if $d(\theta, \theta^\dagger) \le r$ with $r < r_0$, then for every $(\nu, x) \in K \times \mathcal{X}$,

$$\delta_x(\nu, \theta, \mathcal{Q}) \le \delta_x(\nu, \theta^\dagger, \mathcal{Q}) + 2\,\omega(r). \quad (6)$$

By definition of ω, and since $K \subset U$, for every $\nu' \in K$

$$\mathcal{L}_x(\nu', \theta, \mathcal{Q}) \le \mathcal{L}_x(\nu', \theta^\dagger, \mathcal{Q}) + \omega(r) \le \sup_{\nu'' \in U} \mathcal{L}_x(\nu'', \theta^\dagger, \mathcal{Q}) + \omega(r).$$

Take the supremum over $\nu' \in K$; by (H3) at θ, it is also the supremum over U:

$$\sup_{\nu' \in U} \mathcal{L}_x(\nu', \theta, \mathcal{Q}) \le \sup_{\nu' \in U} \mathcal{L}_x(\nu', \theta^\dagger, \mathcal{Q}) + \omega(r).$$

Since $\nu \in K$, also $-\mathcal{L}_x(\nu, \theta, \mathcal{Q}) \le -\mathcal{L}_x(\nu, \theta^\dagger, \mathcal{Q}) + \omega(r)$. Adding the two inequalities gives (6). We apply (6) with $\theta = \hat\theta_F$, $x = x_n$ and $\nu = f_0(x_n)$, which lies in K. On the event $\{d(\hat\theta_F, \theta^\dagger) < r\}$,

$$\frac{1}{N} \sum_{n=1}^{N} \delta_n\big(f_0(x_n), \hat\theta_F, \mathcal{Q}\big) \le \frac{1}{N} \sum_{n=1}^{N} \delta_{x_n}\big(f_0(x_n), \theta^\dagger, \mathcal{Q}\big) + 2\,\omega(r). \quad (7)$$

Step 3: a quantitative law of large numbers at $\theta^\dagger$. Since f_0 and $\theta^\dagger$ are fixed, $\delta_{x_n}\big(f_0(x_n), \theta^\dagger, \mathcal{Q}\big)$ depends on x_n only. By (H1) these N variables are i.i.d. They are measurable, by (H5) and

the measurability of f_0; they take values in $[0, B]$; and their mean is $\mathbb{E}_{X\sim P}\big[\delta_X\big(f_0(X), \theta^\dagger, \mathcal{Q}\big)\big]$. By Hoeffding's inequality (Hoeffding, 1963), with probability at least $1-\alpha$,

$$\frac{1}{N}\sum_{n=1}^{N} \delta_{x_n}\big(f_0(x_n), \theta^\dagger, \mathcal{Q}\big) \le \mathbb{E}_{X\sim P}\big[\delta_X\big(f_0(X), \theta^\dagger, \mathcal{Q}\big)\big] + B\sqrt{\frac{\log(1/\alpha)}{2N}}. \tag{8}$$

Step 4: proof of (i). On the intersection of $\{d(\hat\theta_F, \theta^\dagger) < r\}$ and of the event of (8), the inequalities (5), (7) and (8) chain into (3). By the union bound, the complement of this intersection has probability at most $\Pr\big(d(\hat\theta_F, \theta^\dagger) \ge r\big) + \alpha$.

Step 5: proof of (ii). Let $\varepsilon > 0$. Since $\mathcal{F}$ is nonempty, $0 \le \mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) \le B < \infty$. By definition of the infimum, there is $f_0 \in \mathcal{F}$ with $\mathbb{E}\big[\delta_X\big(f_0(X), \theta^\dagger, \mathcal{Q}\big)\big] \le \mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) + \varepsilon/3$. This f_0 depends on P, $\theta^\dagger$, $\mathcal{F}$ and $\mathcal{Q}$ but not on the data, so (i) applies. Take r as in the statement and $\alpha = \exp\big(-2N\varepsilon^2/(9B^2)\big)$, so that $B\sqrt{\log(1/\alpha)/(2N)} = \varepsilon/3$. The right-hand side of (3) is then at most $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) + \varepsilon$, which gives (4). The first term of the bound tends to 0 by (H2); the second decays exponentially in N.

Step 6: proof of (iii). By Corollary 2, for every dataset, $\mathcal{G}_{\mathrm{am}}(\hat\theta_A, \mathcal{F}, \mathcal{Q}) \le \mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) \le \mathcal{G}_{\mathrm{am}}(\hat\theta_F, \mathcal{F}, \mathcal{Q})$. An upper bound on $\mathcal{G}_{\mathrm{am}}(\hat\theta_F, \mathcal{F}, \mathcal{Q})$ that holds on an event therefore holds on the same event for the two other gaps. □

3.5 Rate and fixed decoder

Corollary 4 (rate). *Let $\mathcal{F}$ be as in Theorem 3. Assume (H1)–(H5), that $\omega(r) \le \kappa r$ for $r \in (0, r_0)$ and some constant κ, that $\sqrt{N}\, d(\hat\theta_F, \theta^\dagger)$ is bounded in probability, and that the infimum defining $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q})$ is attained. For every $\alpha \in (0,1)$ there are s_α and N_α, which do not depend on N, such that for $N \ge N_\alpha$*

$$\Pr\Big(\frac{\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q})}{N} > \mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) + \frac{2\kappa s_\alpha + B\sqrt{\log(1/\alpha)/2}}{\sqrt{N}}\Big) \le 2\alpha.$$

In particular $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) \le N\,\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) + O_P(\sqrt{N})$: the global gap grows at most linearly in N, with slope at most $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q})$.

Proof. Boundedness in probability gives s_α with $\Pr\big(\sqrt{N}\, d(\hat\theta_F, \theta^\dagger) \ge s_\alpha\big) \le \alpha$ for every N. Take N_α with $s_\alpha/\sqrt{N_\alpha} < r_0$, let f_0 attain the infimum, and apply Theorem 3(i) and (iii) with $r = s_\alpha/\sqrt{N}$. Then $\Pr\big(d(\hat\theta_F, \theta^\dagger) \ge r\big) \le \alpha$, $2\,\omega(r) \le 2\kappa s_\alpha/\sqrt{N}$, and $B\sqrt{\log(1/\alpha)/(2N)} = B\sqrt{\log(1/\alpha)/2}/\sqrt{N}$. □

The condition $\omega(r) \le \kappa r$ holds, for example, for an MLP decoder with a Gaussian or Bernoulli likelihood, a Gaussian family $\mathcal{Q}$, and bounded K and $\mathcal{X}$: the reconstruction term is then Lipschitz in θ near $\theta^\dagger$, with a constant integrable under $q(\cdot, \nu)$ for $\nu \in K$. The rate $d(\hat\theta_F, \theta^\dagger) = O_P(N^{-1/2})$ is the parametric rate of M-estimators (Remark 5).

Corollary 5 (fixed decoder). *Let $\Theta = \{\theta\}$, with θ chosen independently of the data, and let $\mathcal{F}$ be a nonempty class of measurable maps $\mathcal{X} \to K$. Assume (H1), and (H3) and (H5) with $\theta^\dagger = \theta$. Then, for every $N \ge 1$ and every $\varepsilon > 0$,*

$$\Pr\Big(\frac{\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})}{N} > \mathcal{R}(\theta, \mathcal{F}, \mathcal{Q}) + \varepsilon\Big) \le \exp\Big(-\frac{N\varepsilon^2}{2B^2}\Big),$$

and the right-hand side improves to $\exp(-2N\varepsilon^2/B^2)$ if the infimum defining $\mathcal{R}(\theta, \mathcal{F}, \mathcal{Q})$ is attained.

Proof. With $\Theta = \{\theta\}$, $\hat\theta_F = \theta$, so (H2) holds with $\theta^\dagger = \theta$, $\omega \equiv 0$ and $\Pr\big(d(\hat\theta_F, \theta^\dagger) \ge r\big) = 0$. In Theorem 3(i), take f_0 with $\mathbb{E}\big[\delta_X\big(f_0(X), \theta, \mathcal{Q}\big)\big] \le \mathcal{R}(\theta, \mathcal{F}, \mathcal{Q}) + \varepsilon/2$ and $\alpha = \exp\big(-N\varepsilon^2/(2B^2)\big)$, so that $B\sqrt{\log(1/\alpha)/(2N)} = \varepsilon/2$. If the infimum is attained, take f_0 attaining it and $\alpha = \exp(-2N\varepsilon^2/B^2)$. □

This is the setting of the frozen-decoder experiment of Cremer et al. (2018, Sec. 5.3), up to one caveat discussed in Section 5.1: the decoder must be independent of the data on which the encoders are retrained.

3.6 Remarks

Remark 4 (why the size of $\mathcal{F}$ does not enter). The proof uses a single encoder f_0, chosen without looking at the data, and applies the law of large numbers to that single function. No uniform law over $\mathcal{F}$ is needed, hence no condition on size, continuity or weight norms. The order of the steps matters: Hoeffding's inequality cannot be applied to $\delta_n\big(f_0(x_n), \hat{\theta}_F, \mathcal{Q}\big)$ directly, since $\hat{\theta}_F$ depends on all the data, so Step 2 first replaces $\hat{\theta}_F$ by the deterministic $\theta^\dagger$.

On the other side, by definition of the infimum, every $f \in \mathcal{F}$ has an expected gap on new data $\mathbb{E}\big[\delta_X\big(f(X), \theta^\dagger, \mathcal{Q}\big)\big] \geq \mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q})$ at $\theta^\dagger$. Asymptotically, $\mathcal{R}$ therefore sits between the training gap of the best encoder (below) and the gap on new data of every encoder of the class (above). The reverse inequality for the training gap, $\mathcal{G}_{\mathrm{am}}(\hat{\theta}_F, \mathcal{F}, \mathcal{Q})/N \geq \mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) - \varepsilon$, would need a uniform law over $\mathcal{F}$. It can fail for a class that interpolates the targets on the dataset without approximating the limiting function elsewhere: this is overfitting of the encoder.

Remark 5 (what $\theta^\dagger$ is). By (2), $\mathcal{L}_F(\theta, \mathcal{Q})/N = \frac{1}{N}\sum_n \sup_{\nu \in U} \mathcal{L}_{x_n}(\nu, \theta, \mathcal{Q})$ is an empirical mean, so $\hat{\theta}_F$ is an M-estimator. The natural candidate for $\theta^\dagger$ is a maximizer of the population F-VI objective

$$\theta \mapsto \mathbb{E}_{X\sim P}\Big[\sup_{\nu\in U} \mathcal{L}_X(\nu, \theta, \mathcal{Q})\Big] = \mathbb{E}_{X\sim P}\big[\log p_\theta(X)\big] - \mathbb{E}_{X\sim P}\Big[\inf_{\nu\in U} \mathrm{KL}\big(q(\cdot, \nu, \mathcal{Q}) \,\|\, p_\theta(\cdot \mid X)\big)\Big].$$

It is the limit of $\mathcal{L}_F(\theta, \mathcal{Q})/N = \big(\log p_\theta(x_{1:N}) - \mathcal{G}_{\mathrm{ap}}(\theta, \mathcal{Q})\big)/N$, where $\mathcal{G}_{\mathrm{ap}}(\theta, \mathcal{Q}) \coloneqq \log p_\theta(x_{1:N}) - \mathcal{L}_F(\theta, \mathcal{Q})$ is the approximation gap. (H2) is then the consistency of M-estimators: uniform convergence of the criterion and a well-separated maximum (van der Vaart, 1998, Thm. 5.7). The rate $d(\hat{\theta}_F, \theta^\dagger) = O_P(N^{-1/2})$ of Corollary 4 holds under the conditions of van der Vaart (1998, Thm. 5.23).

$\theta^\dagger$ depends on Θ, P and $\mathcal{Q}$, but not on $\mathcal{F}$, since F-VI uses no encoder. In general it is neither the data-generating parameter nor the limit of $\hat{\theta}_{\mathrm{ML}}$: it trades likelihood for a smaller approximation gap, which is the bias of variational EM (Turner and Sahani, 2011). The proof does not need it to be either. The decoder $\hat{\theta}_A$ of the VAE has its own limit, which depends on $\mathcal{F}$.

Remark 6 (when $\theta^\dagger$ exists; a version in function space). Apply van der Vaart (1998, Thm. 5.7) to the criterion $\theta \mapsto \sup_\nu \mathcal{L}_x(\nu, \theta, \mathcal{Q})$, with three conditions: a compact Θ (bounded weights, a decoder variance bounded below), continuity in θ for each x, and an integrable envelope. They give a nonempty compact set $\Theta^\dagger$ of population maximizers with $\inf_{\theta\in\Theta^\dagger} d(\hat{\theta}_F, \theta) \to 0$ in probability. This is convergence to a set, and to a point only if the maximizer is unique. Three things can fail.

(a) The supremum can be infinite. For Gaussian decoders with learned variance, the likelihood of deep latent variable models is unbounded (Mattei and Frellsen, 2018), and the F-VI ELBO, whose local factors are free to concentrate, can be unbounded as well.

(b) For a non-compact Θ, the supremum may be approached only by diverging weights.

(c) Maximizers need not be unique. Permutations of hidden units give finitely many copies, and ReLU rescalings a continuum, all defining the same decoder; $\hat{\theta}_F$ may then wander in weight space while $p_{\hat{\theta}_F}(x \mid z)$ converges.

The proof uses $\hat{\theta}_F$ only through the functions $(\nu, x) \mapsto \mathcal{L}_x(\nu, \hat{\theta}_F, \mathcal{Q})$, so (H2) and (H4) can be replaced by

(H2′) there is a fixed function $\ell^\dagger$ on $K \times \mathcal{X}$ with $\sup_{(\nu,x)\in K\times\mathcal{X}} \left|\mathcal{L}_x(\nu, \hat{\theta}_F, \mathcal{Q}) - \ell^\dagger(\nu, x)\right| \to 0$ in probability.

The proof then goes through with three changes:

- $\delta_x(\nu, \theta^\dagger, \mathcal{Q})$ is replaced by $\delta^\dagger(\nu, x) \coloneqq \sup_{\nu'\in K} \ell^\dagger(\nu', x) - \ell^\dagger(\nu, x)$ (in $\mathcal{R}$ and in B), and $2\,\omega(r)$ by twice the supremum in (H2′);
- (H3) is required at $\hat{\theta}_F$ only, with probability tending to one;
- (H5) is written for $\delta^\dagger$.

Symmetries then disappear. Moreover, $\ell^\dagger$ may be the limit of ELBOs of decoders of Θ without being attained in Θ, provided it is finite. Suppose instead that $\hat{\theta}_F$ oscillates between finitely many maximizers that define different functions, with (H3)–(H5) at each of them. A union bound over these maximizers in Steps 2–3 then gives the result with $\max_{\theta\in\Theta^\dagger} \mathcal{R}(\theta, \mathcal{F}, \mathcal{Q})$ in place of $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q})$. A continuum of such maximizers would need a uniform law over $\Theta^\dagger$, which we have not written.

Remark 7 (a fixed width suffices for any accuracy). Suppose in addition that:

- $\mathcal{X} \subset \mathbb{R}^d$ is compact and K is convex;
- there is a measurable $g^\dagger : \mathcal{X} \to K$ with $g^\dagger(x) \in \arg\max_{\nu\in K} \mathcal{L}_x(\nu, \theta^\dagger, \mathcal{Q})$ (one exists by the measurable maximum theorem, Aliprantis and Border, 2006, Thm. 18.19, when $\mathcal{L}_x$ is continuous and K compact);
- $\delta_x(\nu, \theta^\dagger, \mathcal{Q}) \le \frac{\beta}{2}\|\nu - g^\dagger(x)\|^2$ on $K \times \mathcal{X}$ for a constant β. This holds if $\nabla_\nu \mathcal{L}_x(\cdot, \theta^\dagger, \mathcal{Q})$ is β-Lipschitz on K and $g^\dagger(x)$ is interior.

Let $\mathcal{F}_k$ be the one-hidden-layer networks of width k with a continuous non-polynomial activation, followed by the Euclidean projection Π_K onto K. Three facts combine:

- continuous functions are dense in $L^2(P)$;
- such networks are dense in $C(\mathcal{X})$ (Leshno et al., 1993);
- Π_K is non-expansive, with $\Pi_K g^\dagger = g^\dagger$.

Hence $\mathcal{R}(\theta^\dagger, \mathcal{F}_k, \mathcal{Q}) \le \frac{\beta}{2} \inf_f \|f - g^\dagger\|^2_{L^2(P)} \to 0$ as $k \to \infty$. With Theorem 3, for every $\varepsilon > 0$ there is a width $k(\varepsilon)$, independent of N, with $\Pr\left(\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}_{k(\varepsilon)}, \mathcal{Q})/N > \varepsilon\right) \to 0$ (working notes v5, Cor. 5). The map $g^\dagger$ need not be continuous. Section 4 shows that its discontinuities are paid in Lipschitz constant rather than in width.

Remark 8 (Bayesian global factor). Margossian and Blei (2024) do full Bayesian inference over θ. With a global factor q_0 on Θ, the ELBO of the model of Section 1 is

$$-\mathrm{KL}\big(q_0 \,\|\, p(\theta)\big) + \sum_{n=1}^{N} \int_\Theta \mathcal{L}_n(\nu_n, \theta, \mathcal{Q})\, q_0(d\theta).$$

The first term does not depend on the ν_n and cancels in the per-datapoint suboptimalities. The global gap is defined with suprema over q_0, and the analogue of Corollary 2 gives $\mathcal{G}_{\mathrm{am}} \le \inf_{f\in\mathcal{F}} \sum_n \delta_n\big(f(x_n), q_0^*\big)$. Here q_0^* is the F-VI optimal global factor, and $\delta_n(\nu, q_0^*)$ is the suboptimality of $\nu \mapsto \int \mathcal{L}_n(\nu, \theta, \mathcal{Q})\, q_0^*(d\theta)$. The proof is the same with q_0^* in place of $\hat{\theta}_F$ and with (H3) written for $\nu \mapsto \int \mathcal{L}_x(\nu, \theta, \mathcal{Q})\, q_0^*(d\theta)$. Only Step 2 changes, because q_0^* is not a point:

$$\Big|\int_\Theta \mathcal{L}_x(\nu, \theta, \mathcal{Q})\, q_0^*(d\theta) - \mathcal{L}_x(\nu, \theta^\dagger, \mathcal{Q})\Big| \le \omega(r) + \int_{d(\theta,\theta^\dagger)>r} \left|\mathcal{L}_x(\nu, \theta, \mathcal{Q}) - \mathcal{L}_x(\nu, \theta^\dagger, \mathcal{Q})\right| q_0^*(d\theta).$$

One therefore needs an envelope: a measurable $M \geq 0$ on Θ with $\sup_{K\times\mathcal{X}} |\mathcal{L}_x(\nu,\theta,\mathcal{Q}) - \mathcal{L}_x(\nu,\theta^\dagger,\mathcal{Q})| \leq M(\theta)$ and $\int_{d(\theta,\theta^\dagger)>r} M\, dq_0^* \to 0$ in probability for every $r > 0$. This is concentration of q_0^* at $\theta^\dagger$ with integrability; Wang and Blei (2019, Thm. 5) give the concentration of mean-field variational posteriors in regular models. The term $2\,\omega(r)$ in (3) becomes $2\,\omega(r) + 2\int_{d(\theta,\theta^\dagger)>r} M\, dq_0^*$.

4 A jump in the optimal map costs Lipschitz constant

Section 3 shows that the size of the encoder need not grow with N. A discontinuous optimal map has another cost: the encoder must be steep. Throughout this section the decoder is fixed, $\Theta = \{\theta\}$, as in the frozen-decoder experiment of Cremer et al. (2018, Sec. 5.3). By Lemma 1(i), $\mathcal{G}_{\mathrm{am}}(\theta,\mathcal{F},\mathcal{Q}) = \inf_{f\in\mathcal{F}} \sum_n \delta_n\big(f(x_n),\theta,\mathcal{Q}\big)$, and $\delta_x(\nu,\theta,\mathcal{Q})$ is the excess KL divergence of Section 3.

4.1 Separated basins

Scalar input and Lipschitz encoders. The input is scalar, $\mathcal{X} \subset \mathbb{R}$: the crossing argument below works on intervals, and higher-dimensional inputs are handled line by line (Remark 12). $\|\cdot\|$ is the Euclidean norm on U. An encoder $f : \mathcal{X} \to K$ is Λ-Lipschitz if $\|f(x) - f(x')\| \leq \Lambda|x - x'|$ for all $x, x' \in \mathcal{X}$. Λ is the maximal slope of the encoder, the quantity the theorem constrains; for a ReLU network, it is at most the product of the operator norms of the weight matrices.

Definition 3 (separated basins)**.** Let $x_c \in \mathcal{X}$, $r > 0$, $I = (x_c - r, x_c + r) \subset \mathcal{X}$, $I_- = (x_c - r, x_c)$ and $I_+ = (x_c, x_c + r)$. Let $W_-, W_+ \subset K$ be closed with $\|\nu - \nu'\| \geq D > 0$ for all $\nu \in W_-$, $\nu' \in W_+$. Call $K \setminus (W_- \cup W_+)$, assumed nonempty, the *corridor*. The *barrier* and the *wrong-basin excess* are

$$\eta_B \coloneqq \inf_{x\in I} \inf_{\nu\in K\setminus(W_-\cup W_+)} \delta_x(\nu,\theta,\mathcal{Q}), \qquad e(x) \coloneqq \begin{cases} \inf_{\nu\in W_+} \delta_x(\nu,\theta,\mathcal{Q}), & x \in I_-, \\ \inf_{\nu\in W_-} \delta_x(\nu,\theta,\mathcal{Q}), & x \in I_+. \end{cases}$$

The basins are *separated on* I if $\eta_B > 0$.

I localizes the jump at x_c. Typically, the optimal parameters lie in W_- on I_- and in W_+ on I_+, and D is the distance the encoder must travel from one to the other. η_B is the smallest excess KL divergence in the corridor over the whole interval, and $e(x)$ is the smallest excess in the basin that is not that of the side of x. These are the two costs between which a Lipschitz encoder must choose: cross the corridor, or stay in one basin. In the example of Section 4.5, $W_\pm = \{(\mu, \log\sigma_q) : \pm\mu \geq 1\}$ are the two modes and $D = 2$.

Remark 9 (no continuous selection). Assume that $\eta_B > 0$, that $e > 0$ on $I_- \cup I_+$, and that the per-datapoint optimum is attained in K. The optimum has zero excess, so it lies neither in the corridor nor in the wrong basin: it is in W_- on I_- and in W_+ on I_+. Since I is connected and W_-, W_+ are disjoint closed sets, no selection of the argmax is continuous on I. If δ is continuous and K is compact, the argmax correspondence is upper hemicontinuous (Aliprantis and Border, 2006, Thm. 17.31). Selections are then continuous wherever the argmax is a singleton, and jumps occur only at ties (see also Bonnans and Shapiro, 2000). Definition 3 measures how high the barrier between the tied optima is (η_B) and what it costs to sit in the wrong one (e). Theorem 7 turns the topological fact into a cost for Lipschitz selections.

Assumption (R). K is compact and $(\nu, x) \mapsto \delta_x(\nu,\theta,\mathcal{Q})$ is finite and continuous on $K \times \mathcal{X}$. It ensures that e is continuous on $I_\pm$, as the infimum of a continuous function over a compact set. It also ensures that $x \mapsto \delta_x(f(x),\theta,\mathcal{Q})$ is continuous for Lipschitz f, so that all expectations below are defined. It serves no other purpose.

Two population quantities.

$$c_P(\lambda) \coloneqq \inf\big\{P(J):\ J \subset I \text{ an open interval of length } \lambda\big\} \quad \big(c_P(\lambda) \coloneqq +\infty \text{ if } \lambda \geq 2r\big),$$

$$\Gamma_{\pm} \coloneqq \mathbb{E}_{X\sim P}\big[\min\big(\eta_B, e(X)\big)\,\mathbf{1}_{I_{\pm}}(X)\big].$$

$c_P(\lambda)$ is the smallest probability of an interval of length λ in I; it converts a length of inputs into probability without assuming a density. $\Gamma_{\pm}$ is the expected cost of an encoder that stays on the wrong side on $I_{\pm}$.

4.2 The crossing lemma

Lemma 6 (crossing)**.** *Let $J \subset \mathbb{R}$ be an interval and $f : J \to K$ be Λ-Lipschitz. If $f(J)$ meets both W_- and W_+, then the set of $x \in J$ with $f(x)$ in the corridor contains an open interval of length at least D/Λ.*

Proof. Let $S_- = \{x \in J : f(x) \in W_-\}$ and $S_+ = \{x \in J : f(x) \in W_+\}$. They are disjoint, since $W_- \cap W_+ = \emptyset$, and closed in J, since f is continuous and $W_{\pm}$ are closed. Pick $s \in S_-$ and $s' \in S_+$, and assume $s < s'$ (the other case is symmetric).

- $S_- \cap [s, s']$ is compact and contains s, so it has a largest element u, and $u < s'$ because $s' \notin S_-$.
- $S_+ \cap [u, s']$ is compact and contains s', so it has a smallest element v, and $v > u$ because $u \notin S_+$.

A point $t \in (u, v)$ lies in $[s, s']$; it is not in S_- since $t > u$, and not in S_+ since $u < t < v$. Hence $f(t)$ is in the corridor for every $t \in (u, v)$. Finally, $f(u) \in W_-$ and $f(v) \in W_+$, so $D \leq \|f(v) - f(u)\| \leq \Lambda(v - u)$. □

This is the intermediate value theorem made quantitative: a continuous path from one basin to the other must pass through the corridor, and a Λ-Lipschitz path spends an input length at least D/Λ there.

4.3 Statement

Theorem 7 (two-threshold bound)**.** *Assume (R) and that the basins are separated on I.*

(i) (population) *Every Λ-Lipschitz encoder $f : \mathcal{X} \to K$ satisfies*

$$\mathbb{E}_{X\sim P}\big[\delta_X\big(f(X), \theta, \mathcal{Q}\big)\big] \ \geq\ \min\big(\eta_B\, c_P(D/\Lambda),\ \Gamma_-,\ \Gamma_+\big). \tag{9}$$

In particular, $\mathcal{R}(\theta, \mathcal{F}, \mathcal{Q})$ has the same lower bound for every class $\mathcal{F}$ of Λ-Lipschitz encoders.

(ii) (data) *Let $x_1, \ldots, x_N$ be i.i.d. with distribution P, $\alpha \in (0, 1)$ and $\varepsilon_N \coloneqq \sqrt{\log(4/\alpha)/(2N)}$. With probability at least $1 - \alpha$, simultaneously for every class $\mathcal{F}$ of Λ-Lipschitz encoders (possibly depending on N or on the data),*

$$\frac{\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})}{N} \ \geq\ \min\Big(\eta_B\big(c_P(D/\Lambda) - 2\varepsilon_N\big),\ \Gamma_- - \eta_B\varepsilon_N,\ \Gamma_+ - \eta_B\varepsilon_N\Big). \tag{10}$$

Moreover, almost surely, $\liminf_{N\to\infty} \mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})/N \ \geq\ \min\big(\eta_B\, c_P(D/\Lambda), \Gamma_-, \Gamma_+\big)$ for every such class.

(iii) (density) *If P has a density at least $\underline{p} > 0$ on I, then $c_P(\lambda) \geq \underline{p}\,\lambda$ for $\lambda < 2r$ and $\Gamma_{\pm} \geq \underline{p} \int_{I_{\pm}} \min\big(\eta_B, e(x)\big)\, dx$. In particular,*

$$\mathbb{E}_{X\sim P}\big[\delta_X\big(f(X), \theta, \mathcal{Q}\big)\big] \ \geq\ \min\Big(\frac{\underline{p}\,\eta_B\, D}{\Lambda},\ \Gamma_-,\ \Gamma_+\Big). \tag{11}$$

Reading. On I, a Λ-Lipschitz encoder has two options.

- It can *cross*. It then passes through the corridor on an interval of inputs of length at least D/Λ, and pays at least η_B per datapoint there: the first term, about $\underline{p}\,\eta_B D/\Lambda$.
- It can *stay*. It remains on one side on all of I, and pays at least $\min(\eta_B, e(x))$ on the half where its basin is the wrong one: the terms $\Gamma_\pm$.

The theorem says that it cannot do better than the cheaper of the two. By (11), reaching a per-datapoint gap $\varepsilon < \min(\Gamma_-, \Gamma_+)$ requires

$$\Lambda \;\geq\; \frac{\underline{p}\,\eta_B\, D}{\varepsilon}.$$

Combined with Section 3 (Corollary 5 and Remark 7), the size of the encoder need not grow with N, but if the optimal map jumps, its Lipschitz constant must grow as $1/\varepsilon$. For a class of Λ-Lipschitz encoders, both the population gap $\mathcal{R}(\theta, \mathcal{F}, \mathcal{Q})$ and the per-datapoint training gap $\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})/N$ are bounded below by $\min\big(\eta_B c_P(D/\Lambda), \Gamma_-, \Gamma_+\big)$, the latter up to $O(\eta_B \varepsilon_N)$. Under (H3), Corollary 5 bounds the same training gap above by $\mathcal{R}(\theta, \mathcal{F}, \mathcal{Q}) + \varepsilon$, with probability at least $1 - \exp\big(-N\varepsilon^2/(2B^2)\big)$.

4.4 Proof

Proof of Theorem 7. Step 1: a pointwise lower bound for an arbitrary encoder. Fix a Λ-Lipschitz encoder f. By definition of η_B and e, for every $x \in I$:

- if $f(x)$ is in the corridor, then $\delta_x(f(x), \theta, \mathcal{Q}) \geq \eta_B$;
- if $x \in I_+$ and $f(x) \in W_-$, or $x \in I_-$ and $f(x) \in W_+$, then $\delta_x(f(x), \theta, \mathcal{Q}) \geq e(x)$.

Outside I we only use $\delta \geq 0$. At least one of the following three cases occurs.

(A) *$f(I)$ meets both W_- and W_+.* Lemma 6, applied to the restriction of f to I, gives an open interval $J \subset I$ of length D/Λ on which f is in the corridor. Hence, for every $x \in \mathcal{X}$,

$$\delta_x\big(f(x), \theta, \mathcal{Q}\big) \geq \eta_B\, \mathbf{1}_J(x). \tag{A}$$

This case is impossible if $D/\Lambda \geq 2r$: the interval (u, v) of the lemma has its endpoints in I, hence length less than $2r$. This is the reason for the convention $c_P(\lambda) = +\infty$ for $\lambda \geq 2r$.

(B) *$f(I)$ does not meet W_+.* For $x \in I_+$, $f(x)$ is then in W_- or in the corridor, so

$$\delta_x\big(f(x), \theta, \mathcal{Q}\big) \geq \min\big(\eta_B, e(x)\big)\, \mathbf{1}_{I_+}(x). \tag{B}$$

(C) *$f(I)$ does not meet W_-.* Symmetrically,

$$\delta_x\big(f(x), \theta, \mathcal{Q}\big) \geq \min\big(\eta_B, e(x)\big)\, \mathbf{1}_{I_-}(x). \tag{C}$$

Case (A) is the encoder that crosses; (B) and (C) are the encoders that stay on one side.

Step 2: proof of (i). By (R), $x \mapsto \delta_x(f(x), \theta, \mathcal{Q})$ is continuous and e is continuous on $I_\pm$, so the expectations are defined. We integrate Step 1 under P.

- In case (A), $\mathbb{E}\big[\delta_X(f(X), \theta, \mathcal{Q})\big] \geq \eta_B P(J) \geq \eta_B\, c_P(D/\Lambda)$, since $J \subset I$ is an open interval of length D/Λ.
- In case (B) the expectation is at least Γ_+, and in case (C) at least Γ_-.

The statement on $\mathcal{R}(\theta, \mathcal{F}, \mathcal{Q})$ follows by taking the infimum over $f \in \mathcal{F}$.

Step 3: proof of (ii). The encoder that attains the infimum in $\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})$ depends on the data, so the law of large numbers cannot be applied to a fixed function, as in Theorem 3. But Step 1 involves only three kinds of functions: indicators $\mathbf{1}_J$ of intervals, where J depends on f, and the two fixed functions $\min(\eta_B, e)\,\mathbf{1}_{I_\pm}$. It suffices to control empirical means uniformly over intervals, and for two fixed functions. Write $P_N(A) := \frac{1}{N}\#\{n \le N : x_n \in A\}$ for the empirical distribution.

- *Intervals.* By the Dvoretzky–Kiefer–Wolfowitz inequality with Massart's constant (Dvoretzky et al., 1956; Massart, 1990),

$$\Pr\Big(\sup_{t\in\mathbb{R}} \big|P_N((-\infty,t]) - P((-\infty,t])\big| > \varepsilon_N\Big) \le 2\exp(-2N\varepsilon_N^2) = \alpha/2.$$

  On the complement, write $P_N(J) = P_N((-\infty,b)) - P_N((-\infty,a])$ for every open interval $J = (a,b)$, and likewise for P. Each term is within ε_N of its counterpart (for $(-\infty,b)$, by letting $t \uparrow b$). Hence $P_N(J) \ge P(J) - 2\varepsilon_N$ for every open interval J.

- *Two fixed functions.* The variables $\min(\eta_B, e(x_n))\,\mathbf{1}_{I_+}(x_n)$ are i.i.d., take values in $[0,\eta_B]$, and have mean Γ_+. By Hoeffding's inequality, with probability at least $1 - \exp\big(-2N(\eta_B\varepsilon_N)^2/\eta_B^2\big) = 1 - \alpha/4$,

$$\frac{1}{N}\sum_{n=1}^{N} \min\big(\eta_B, e(x_n)\big)\,\mathbf{1}_{I_+}(x_n) \ge \Gamma_+ - \eta_B\varepsilon_N,$$

  and likewise for I_- and Γ_-.

These three events involve no encoder, and their intersection has probability at least $1-\alpha$. On it, take any Λ-Lipschitz encoder f and average Step 1 over the data.

- In case (A), $\frac{1}{N}\sum_n \delta_n(f(x_n),\theta,\mathcal{Q}) \ge \eta_B P_N(J) \ge \eta_B\big(P(J) - 2\varepsilon_N\big) \ge \eta_B\big(c_P(D/\Lambda) - 2\varepsilon_N\big)$.
- In case (B) the average is at least $\Gamma_+ - \eta_B\varepsilon_N$, and in case (C) at least $\Gamma_- - \eta_B\varepsilon_N$.

The lower bound thus holds for all Λ-Lipschitz encoders at once. By Lemma 1(i), the gap $\mathcal{G}_{\mathrm{am}}(\theta,\mathcal{F},\mathcal{Q})/N$ is an infimum of such averages, which gives (10). Since the event does not depend on $\mathcal{F}$, the measurability of $\mathcal{G}_{\mathrm{am}}$ plays no role.

Step 4: the almost-sure statement. By the Glivenko–Cantelli theorem, $\sup_t |P_N((-\infty,t]) - P((-\infty,t])| \to 0$ almost surely, and by the strong law of large numbers the two empirical means converge to $\Gamma_\pm$. On the intersection of these probability-one events, the argument of Step 3, with these vanishing deviations in place of ε_N, gives the bound on the limit inferior.

Step 5: proof of (iii). If P has a density at least $\underline{p}$ on I, every open interval $J \subset I$ of length $\lambda < 2r$ has $P(J) \ge \underline{p}\lambda$, and $\Gamma_\pm = \int_{I_\pm} \min(\eta_B, e(x))\,P(dx) \ge \underline{p}\int_{I_\pm} \min(\eta_B, e(x))\,dx$. For (11), there are two cases. If $D/\Lambda < 2r$, then $\eta_B\, c_P(D/\Lambda) \ge \underline{p}\,\eta_B D/\Lambda$. If $D/\Lambda \ge 2r$, then $c_P = +\infty$ and the minimum in (9) equals $\min(\Gamma_-,\Gamma_+)$, which is at least the minimum in (11). □

4.5 An example

Take $z \sim \mathcal{N}(0,1)$, $x_1 \mid z \sim \mathcal{N}(z^2, 0.5^2)$, $x_2 \mid z \sim \mathcal{N}(z, 2^2)$, a Gaussian family with $\nu = (\mu, \log\sigma_q) \in K = [-3.5, 3.5] \times [-3, 1]$, and the line $x_1 = 4$, on which the encoder sees x_2. The posterior has two modes near $z = \pm 2$, and x_2 tilts their weights; the optimal Gaussian mean jumps from about -1.95 to about 1.95 at $x_2 = 0$. Take $I = (-1, 1)$, $W_\pm = \{\pm\mu \ge 1\}$ (so $D = 2$), and, for the illustration, x_2 uniform on $(-1, 1)$ (so $\underline{p} = 1/2$). The constants, computed in closed form on grids (working notes v5, App. E.1), are:

- $\eta_B = 13.685$ nats;
- $e(x_2) = 0.974\,|x_2|$;
- $\Gamma_- = \Gamma_+ = 0.2435$.

By (11), every Λ-Lipschitz encoder satisfies

$$\mathbb{E}\,\delta_X\big(f(X),\theta,\mathcal{Q}\big) \;\geq\; \min\Big(\frac{13.685}{\Lambda},\; 0.2435\Big).$$

- *Staying.* For $\Lambda \leq 13.685/0.2435 \approx 56$ the bound is 0.2435, while the best constant encoder (the Gaussian fitted at $x_2 = 0$, with $\mu = \pm 1.948$) achieves 0.2437. On this line, an encoder with Lipschitz constant at most 56 does essentially no better than a constant (within 0.1%).
- *Crossing.* For large Λ, consider the ramp encoder $\mu(x_2) = \max(-\bar{\mu}, \min(\bar{\mu}, \Lambda x_2))$, where $(-\bar{\mu}, \sigma_q) = (-1.952, 0.128)$ is the Gaussian fitted at $x_2 = -1$ and σ_q is kept fixed. It achieves $29.92/\Lambda + 1.7 \cdot 10^{-4}$, so the $1/\Lambda$ coefficient of the bound is within a factor 2.19. The factor is 1.42 with the refinement of Remark 11.

The bound thus has the right form in both regimes. A per-datapoint gap of 0.01 nat requires $\Lambda \geq 1368.5$. On the full input (x_1, x_2), uniform on $[2,6] \times (-1,1)$, the line-by-line version of Remark 12 gives at least 0.240 nat for $\Lambda \leq 5.5$, 0.149 nat at $\Lambda = 100$, and $15.7/\Lambda$ at $\Lambda = 1000$.

4.6 Remarks

Remark 10 (why two thresholds). With a single threshold η, one needs both $\eta \leq \eta_B$ and $e \geq \eta$ away from the jump. If $e \geq \eta$ outside $(x_c - r_1, x_c + r_1)$, the same argument gives $\underline{p}\,\eta \min(D/\Lambda, r - r_1)$. The threshold is then limited by the wrong-basin cost (at most 0.974 nat in the example) instead of the barrier (13.7 nats). With $\eta = 0.5$ and $r_1 = 0.513$, the bound is $0.25 \min(2/\Lambda, 0.487)$, weaker by a factor 27 in the $1/\Lambda$ regime.

Remark 11 (level-set refinement). Let $\psi(\nu) \coloneqq \min\big(\operatorname{dist}(\nu, W_-), D\big)$. This map is 1-Lipschitz, vanishes on W_-, and equals D on W_+. For $c \in (0, D)$, let

$$\tilde{\eta}(c) \coloneqq \inf_{x \in I}\; \inf\big\{\delta_x(\nu,\theta,\mathcal{Q}) : \nu \in K,\; \psi(\nu) = c\big\} \;\geq\; \eta_B.$$

If P has a density at least $\underline{p}$ on I, the term $\eta_B\, c_P(D/\Lambda)$ in (9) can be replaced by $\frac{p}{\Lambda}\int_0^D \tilde{\eta}(c)\,dc$. The proof follows $\psi \circ f$ along the crossing, by the area formula (working notes v5, Remark D.4). Instead of the minimal height of the corridor, the bound then integrates the profile of the barrier along the crossing. In the example, with basins $\{\pm\mu \geq 1.5\}$, the $1/\Lambda$ coefficient becomes 21.04 instead of 13.685.

Remark 12 (several inputs). If $\mathcal{X} \subset \mathbb{R}^d$, write $x = (y,t)$ with $t \in \mathbb{R}$. The restriction of a Λ-Lipschitz encoder to a line $t \mapsto (y,t)$ is Λ-Lipschitz, so Theorem 7(i) applies on each line that crosses the jump set, with constants $\eta_B(y)$, $D(y)$ and $\Gamma_\pm(y)$. Suppose P has a density at least $\underline{p}$ on the union of the intervals. Integrating over y (Fubini) gives

$$\mathbb{E}\,\delta_X\big(f(X),\theta,\mathcal{Q}\big) \geq \underline{p}\int \min\Big(\frac{\eta_B(y)D(y)}{\Lambda}, \Gamma_-(y), \Gamma_+(y)\Big)\,dy,$$

with $\Gamma_\pm(y)$ computed with Lebesgue measure on the line (working notes v5, Cor. D.5). The empirical version needs a uniform law over all Λ-Lipschitz encoders. When $K \times \mathcal{X}$ is compact, it follows from the Arzelà–Ascoli theorem.

Remark 13 (values in U or in a space of distributions). The proof uses only three facts: $W_\pm$ are closed, f is continuous, and δ admits pointwise lower bounds. It therefore holds for encoders with values in U, with η_B and e then infima over larger sets, hence possibly smaller. It also holds

for encoders with values in a set of distributions with a metric (Hellinger, W_2), under measurability conditions (working notes v5, App. D.3). However, Λ depends on the parametrization: a slope in $(\mu, \log\sigma)$ is not a slope in (μ, σ). Comparing families at equal Λ must therefore be done in a space of distributions (working notes v5, Cor. D.6).

Remark 14 (what the theorem does not say). The theorem has three limits.

- It bounds an infimum over a class from below, and says nothing about optimization.
- It does not say whether optimal parameters jump on real data; that is an empirical question. A hysteresis test answers it: optimize per datapoint along a path of inputs $x(t)$, warm-starting from each end, and compare the two solutions in the middle.
- It assumes a fixed decoder. A jointly trained decoder can accommodate, make the posterior unimodal and remove the jump. This is what the planned accommodation experiment can measure.

5 Context and related work

5.1 Theorem 3 in context

Margossian and Blei (2024). They reduce the closure of the global gap, $\mathcal{G}_{\text{am}}(\Theta, \mathcal{F}, \mathcal{Q}) = 0$, to the existence of an ideal inference function, $f(x_n) = \nu_n^*$ for all n (Margossian and Blei, 2024, Def. 2.2, Eq. 5). They then ask "how rich does $\mathcal{F}$ need to be" (Sec. 2) and study the question empirically (Sec. 4). They conclude that the number of parameters of the inference function need not scale with N, whereas that of F-VI must (Sec. 1). Each of their experiments is run at a fixed N ($N = 10{,}000$ in Secs. 4.2–4.4). A small network reaches the F-VI ELBO there (width $k \geq 4$ in Sec. 4.3, $k \geq 64$ in Sec. 4.4), but N is not varied. Theorem 3 gives the argument on the dependence on N, for the per-datapoint gap.

The mechanism is visible in their Eq. 7: the F-VI optimum $q(z_n; \nu_n^*)$ depends on the whole dataset only through the global factor $g(\theta, \mathbf{x}) = q(\theta; \nu_0^*(\mathbf{x}))$, here $\hat{\theta}_F$. By (6), once $\hat{\theta}_F$ is close to $\theta^\dagger$, a maximizer of $\mathcal{L}_x(\cdot, \theta^\dagger, \mathcal{Q})$ in K is, at every x_n, within $2\,\omega\big(d(\hat{\theta}_F, \theta^\dagger)\big)$ of optimal in terms of δ. This function does not depend on N. The encoder does not have to interpolate N arbitrary targets, which would require a class of pseudo-dimension at least N (Bartlett et al., 2019); it only has to approximate this function on average under P.

Suppose $\mathcal{F}$ contains a map f with $f(x) \in \arg\max \mathcal{L}_x(\cdot, \theta^\dagger, \mathcal{Q})$ for P-almost every x. This is an ideal inference function of the limiting problem, defined on the support of P rather than on the N data points. Then $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q}) = 0$ and $\mathcal{G}_{\text{am}}(\Theta, \mathcal{F}, \mathcal{Q})/N \to 0$ in probability. This is weaker than their exact closure, $\mathcal{G}_{\text{am}}(\Theta, \mathcal{F}, \mathcal{Q}) = 0$: Theorem 3 gives $o_P(N)$, and Corollary 4 gives $O_P(\sqrt{N})$.

Three further points:

- The theorem concerns the infimum over $\mathcal{F}$, not what optimization reaches. It says nothing about the seed sensitivity and the convergence failures of A-VI they report (Sec. 4.3).
- In their linear model (Prop. 3.6), an affine encoder closes the gap for every N, consistent with $\mathcal{R} = 0$. The model is not covered as stated: the optimal means $\frac{\tau}{\sigma^2+\tau^2}(x_n - \bar{x})$ (their Eq. 9) are unbounded on $\mathcal{X} = \mathbb{R}$, so (H3) fails.
- Their experiments use a Bayesian global factor, the setting of Remark 8. They also ask how accurate A-VI is on held-out data (Sec. 5). $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q})$ is a held-out quantity: asymptotically, it bounds the training gap from above and the gap on new data of every encoder of $\mathcal{F}$ from below (Remark 4).

Cremer et al. (2018). Their "Amortization" row (Table 2) is $\mathcal{L}[q^*] - \mathcal{L}[q]$, averaged over 1000 training points (100 for CIFAR) at the decoder of the trained VAE. If training reached the A-VI optimum, it estimates $\mathcal{G}_{\mathrm{am}}(\hat{\theta}_A, \mathcal{F}, \mathcal{Q})/N$ up to subsampling error, from below, since their $\mathcal{L}[q^*]$ comes from local optimization. By Theorem 3(iii), this quantity is asymptotically at most $\mathcal{R}(\theta^\dagger, \mathcal{F}, \mathcal{Q})$. Under the assumptions of the theorem, their per-datapoint values for q_{FFG} (1.34 nats on MNIST, 4.22 on Fashion-MNIST, 37.47 on 3-bit CIFAR) therefore need not grow with the size of the training set. The theorem does not say that they are small; that depends on $\mathcal{R}$.

The decrease with a larger encoder (Table 3, left: $1.34 \to 1.11$ and $4.22 \to 3.09$) is consistent with the monotonicity $\mathcal{F} \subset \mathcal{F}' \Rightarrow \mathcal{R}(\theta, \mathcal{F}', \mathcal{Q}) \leq \mathcal{R}(\theta, \mathcal{F}, \mathcal{Q})$, but the two models do not share a decoder. For binarized images, $\mathcal{X} = \{0,1\}^{784}$ is finite, hence compact; the assumption one cannot check for their decoders is (H2).

In their Sec. 5.3 (Table 4), the decoder is frozen and linear encoders are retrained with q_{FFG}, q_{AF} and q_{Flow} (amortization gaps of 43.22, 13.10 and 12.86 nats). This is the case $\Theta = \{\theta\}$ of Corollary 5, with one caveat: the corollary needs a decoder independent of the data on which the encoders are retrained, whereas theirs was trained on the same set. Their three columns change both $\mathcal{F}$ (the flow adds parameters to the encoder, which is their point) and $\mathcal{Q}$. Changing $\mathcal{Q}$ also changes the reference of δ, whose supremum is taken over a richer family (approximation gaps of 1.91, 0.78 and 0.43).

In their Sec. 5.5 (Fig. 3), the amortization gap is larger on validation than on training data, and more so with a larger encoder ("the encoder overfits more than the decoder"). This is the asymmetry of Remark 4. At a fixed decoder, the training gap is bounded above by $\mathcal{R}$ without any condition on $\mathcal{F}$, while the gap on new data of every encoder of $\mathcal{F}$ is at least $\mathcal{R}$. The difference is overfitting of the encoder; controlling it requires a uniform law over $\mathcal{F}$, hence a class of controlled complexity, for instance through Rademacher bounds (Bartlett and Mendelson, 2002).

Finally, the bound goes through the choice $\theta = \hat{\theta}_F$ in Lemma 1(ii), which gives $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) \leq \mathcal{G}_{\mathrm{am}}(\hat{\theta}_F, \mathcal{F}, \mathcal{Q})$. It therefore ignores the accommodation of the decoder (Cremer et al., 2018, Sec. 5.4): the VAE decoder moves to make amortization easier. By Corollary 2, this can only bring $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q})$ and $\mathcal{G}_{\mathrm{am}}(\hat{\theta}_A, \mathcal{F}, \mathcal{Q})$ below $\mathcal{G}_{\mathrm{am}}(\hat{\theta}_F, \mathcal{F}, \mathcal{Q})$. The accommodation gap $\mathcal{G}_{\mathrm{ac}}(\theta) := \log p_{\hat{\theta}_{\mathrm{ML}}}(x_{1:N}) - \log p_\theta(x_{1:N})$ does not enter.

5.2 Theorem 7 in context

Where jumps come from. The reverse divergence $\mathrm{KL}(q \,\|\, p)$ is mode-seeking: a unimodal family fitted to a bimodal posterior locks onto one mode (Minka, 2005; Bishop, 2006, Sec. 10.1.2). Turner and Sahani (2011) discuss the consequences for variational EM. When the observation moves the weights of the two modes through equality, the optimum switches mode at once.

This is a general fact of parametric optimization. The argmax of a nonconvex problem is only upper hemicontinuous (Berge's maximum theorem; Aliprantis and Border, 2006, Thm. 17.31). Selections are continuous where the argmax is a singleton, and can jump only at ties (Bonnans and Shapiro, 2000). In the example, the jump is indeed a tie between two non-degenerate optima, without pitchfork bifurcation (working notes v5, App. E.2). Statistical physics provides the classical instance: in mean-field approximations of ferromagnets below the critical temperature, the magnetization jumps when the external field changes sign. Wainwright and Jordan (2008) discuss the nonconvexity of mean-field variational inference. Definition 3 makes the topological fact "no continuous selection" quantitative, and Theorem 7 turns it into a cost for Lipschitz selections.

The same mechanism for generative models.

- Salmona et al. (2022) consider a Lipschitz push-forward generator (a VAE decoder, a GAN) mapping a Gaussian to a distribution with separated modes. They show that the total variation and the KL divergence are bounded below by a constant that depends on the separation of the modes and on the Lipschitz constant. They deduce a provable trade-off between Lipschitz-based stabilization and multimodality.
- Tanielian et al. (2020) bound from below, by Gaussian isoperimetry, the fraction of GAN samples that fall between disconnected components. Khayatkhoei et al. (2018) show that a single continuous generator cannot cover disconnected manifolds.
- For normalizing flows, Cornish et al. (2020) show that the bi-Lipschitz constant must blow up when the topology of the target differs from that of the base. Hagemann and Neumayer (2021) make the same observation for invertible networks, with mixture latents as a remedy.

Lemma 6 is the same mechanism: a connected domain mapped onto two separated regions forces a corridor of width proportional to $1/\Lambda$. There are three differences.

- *Direction.* Their map goes from latent to data and is judged by a divergence between distributions. Ours goes from data to variational parameters and is judged input by input, by an excess KL divergence.
- *Measure.* The role of the Gaussian measure and of isoperimetry is played by the input distribution P, through c_P.
- *Alternative to crossing.* Theirs is to drop a mode; ours is to stay in the wrong basin. This cost $e(x)$ enters the bound explicitly, which gives the two regimes.

Discontinuous targets and richer families. In regression, a Λ-Lipschitz function approximates a jump of size h with an L^1 error of at least $h^2/(4\Lambda)$ on a long enough interval, and approximations of functions with jumps by ReLU networks rely on steep transitions (e.g. Petersen and Voigtlaender, 2018). Theorem 7 is the variational version. The loss is not a distance to the target but the excess KL divergence, which is at least η_B in the whole corridor, whatever the distance to the basins. The crossing cost, of order $\eta_B D/\Lambda$, is therefore set by the height of the barrier, and is only linear in D.

For multi-valued targets, Bishop (1994) showed that the conditional mean is not a valid target, and proposed mixture density networks. The variational analogue is that a richer family can make the optimal map continuous, and the Lipschitz cost then disappears. On the line of the example, the optimal weight of a two-component mixture follows the posterior mass of $\{z < 0\}$, which varies continuously (working notes v5, App. E.3). This is the same remedy as the mixture latents of Hagemann and Neumayer (2021), or the mixture-density posteriors used in neural posterior estimation (Papamakarios and Murray, 2016). Comparing families at equal Lipschitz constant must be done in a space of distributions, since their parameter spaces differ (Remark 13).

Amortized optimization. Amos (2023) distinguishes regression-based amortization, which fits the solutions, from objective-based amortization, which minimizes the objective at the model output, and discusses non-unique solutions. A model regressed on two optima averages them and lands in the corridor, where it pays at least η_B. A model trained on the objective, as the encoder of a VAE, is free to choose a basin; Theorem 7 bounds from below what it pays when it is Λ-Lipschitz.

Regularity of the encoder: a trade-off. Smoother encoders are sought for other reasons:

- generalization bounds involve spectral norms (Bartlett et al., 2017);

- amortized inference regularization deliberately makes encoders smoother to improve generalization (Shu et al., 2018);
- certifiably robust VAEs rely on Lipschitz constants of the encoder (Barrett et al., 2022).

Theorem 7 quantifies the other side of the trade-off. When the optimal map jumps, lowering Λ costs at least $\underline{p}\,\eta_B D/\Lambda$ per datapoint, down to the regime of the constant encoder. This is the encoder analogue of the trade-off between stability and multimodality of Salmona et al. (2022).

Semi-amortized inference. Hjelm et al. (2016), Kim et al. (2018), Krishnan et al. (2018) and Marino et al. (2018) refine the encoder output by local optimization of the per-datapoint ELBO. The theorem suggests, without proving it, where refinement helps. Near x_c, the encoder outputs lie in the corridor or in the wrong basin.

- From the corridor, a local ascent goes down into one basin, right or wrong.
- From the wrong basin, it stays there if this basin contains a local optimum, as in the example, where both modes are local optima.

Refinement would thus remove the barrier part of the cost, but not necessarily the wrong-basin part. The theorem says nothing about these dynamics.

Cremer et al. (2018).

- Their Fig. 2 (posterior D) shows a bimodal posterior in a VAE with a two-dimensional latent space trained on MNIST. The situation of the theorem therefore occurs on real data.
- In their Sec. 5.3 (Table 4), flows reduce the amortization gap of linear encoders on a frozen decoder from 43.22 to 13.10 and 12.86 nats, which they attribute to the added parameters. Theorem 7 suggests a second possible mechanism. If the optimal factorized-Gaussian parameters jump on their data, every Lipschitz encoder pays the cost of the theorem; a linear encoder is Lipschitz, with constant the operator norm of its matrix. A family whose optimal map is continuous can avoid this cost. Their experiment does not separate the two mechanisms; a hysteresis test would (Remark 14).
- Their Sec. 5.4 (accommodation) indicates that a jointly trained decoder can bring the posterior closer to the family, and so possibly remove jumps. The theorem, at a fixed decoder, does not cover this case.

Margossian and Blei (2024). In a simple hierarchical model, an ideal inference function exists (Margossian and Blei, 2024, Thm. 3.4), but its regularity in x is not discussed. The example of Section 4.5 is a simple hierarchical model with a fixed decoder whose optimal map is discontinuous. Their question "how rich does $\mathcal{F}$ need to be" thus has two axes:

- size, which need not grow with N (Theorem 3, Remark 7);
- steepness, which must grow as $1/\varepsilon$ when the optimal map jumps (Theorem 7).

Their experiments, in which small networks reach the F-VI ELBO at $N = 10{,}000$, do not probe the second axis.

6 Status of the statements and open questions

Table 1 separates what is known from what we did not find in the papers we checked. "Not found" means not found in this form; the search was not exhaustive, and most of these statements are elementary.

Statement	Status	Relies on
Separation (2)	known	Agrawal and Domke (2021, Sec. 4.1, Eq. 8)
Lemma 1	partially known	variational-EM case in Shu et al. (2018, Eqs. 11–12); (i) is the gap of Cremer et al. (2018, Eq. 8) summed over datapoints
Corollary 2	elementary	definitions
Theorem 3	not found	concentration assumed (M-estimation, van der Vaart, 1998; Wang and Blei, 2019 in the Bayesian case); Hoeffding
Corollaries 4, 5	not found (direct)	Theorem 3
Remark 7	not found	universal approximation (Leshno et al., 1993)
Lemma 6	elementary; mechanism known	intermediate value theorem; same mechanism in Salmona et al. (2022)
Theorem 7	not found	Lemma 6; DKW with Massart's constant; Hoeffding

Table 1: Status of the statements and what they rely on.

Open questions.

- A lower bound matching Theorem 3 would require a uniform law over $\mathcal{F}$ and the convergence of $\hat{\theta}_A$; neither is established.
- (H2′) with a continuum of maximizers that define different functions.
- Whether stochastic optimization of the ELBO reaches the infimum over $\mathcal{F}$. The seed sensitivity reported by Margossian and Blei (2024) suggests that it is not automatic.
- Whether optimal Gaussian parameters jump on real data (hysteresis test, Remark 14).
- Lipschitz constants in parameter space versus distribution space, in particular for flow families.
- Whether joint training removes jumps through accommodation, and at which cost in $\mathcal{G}_{\text{ac}}$.

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